Full Revision Booklet

5CM522 Operational Research

Combined and renumbered from Practice Papers A and B

This booklet contains all questions from both current practice papers. The questions have been renumbered so that the set reads as one continuous revision pack.

Booklet Question	Source	Original Question
1–6	Practice Paper A	Questions 1–6
7–12	Practice Paper B	Questions 1–6

1.

Source: Practice Paper A, original Question 1.

Elmbridge Furniture makes two products each week: desks and shelving units.

Each desk uses 5 carpentry hours and 2 finishing hours. Each shelving unit uses 3 carpentry hours and 4 finishing hours. At most 120 carpentry hours and 96 finishing hours are available each week. At least 8 shelving units must be produced to satisfy an online retailer.

The weekly contribution is £58 per desk and £46 per shelving unit.

(a) Formulate a linear programming model for this problem, clearly defining the decision variables, objective function and constraints. (6 Marks)

(b) Solve the model to determine the weekly production plan that maximises contribution. You may use a graphical method, algebra or Python, but your work must be clear. (6 Marks)

(c) Identify the binding constraints and determine the slack in each constraint at the optimum. (4 Marks)

(d) The contribution on shelving units rises to £52. Determine whether the optimal basis changes and justify your answer clearly. (5 Marks)

(e) A new agreement requires shelving units to make up at least 30% of total weekly output. Reformulate the model and determine whether the original optimum remains feasible. If not, find the revised optimal plan. (4 Marks)

Total 25 Marks

2.

Source: Practice Paper A, original Question 2.

GreenGrid Energy is choosing a set of improvement projects for a regional depot network. Each project is either undertaken in full or not undertaken at all.

Project	Cost (£m)	Annual Benefit Score
A	6	14
B	8	17
C	5	11
D	9	20
E	4	9
F	7	15

The total budget is £21 million.

The following additional conditions apply:

- if project D is selected, project B must also be selected;
- projects A and F cannot both be selected;
- at least one of projects C and E must be selected.

(a) Formulate a binary integer programming model to maximise total annual benefit subject to the budget and project-selection rules. Clearly define all decision variables. (8 Marks)

(b) Solve the model in Python or by systematic analysis, and determine which projects should be selected and the maximum total annual benefit score. (7 Marks)

(c) Explain why this problem requires binary decision variables rather than the continuous variables used in ordinary linear programming. (3 Marks)

(d) Modify the model so that exactly four projects must be selected, then resolve or explain how the optimal solution would change. (4 Marks)

(e) Suggest one realistic extension that would make this capital-selection model more useful in practice. No full formulation is required. (3 Marks)

Total 25 Marks

3.

Source: Practice Paper A, original Question 3.

The network below represents proposed communication links between seven depots. The number on each edge is the installation cost in thousands of pounds.

Link	Cost	Link	Cost
A–B	6	C–F	6
A–C	4	D–E	5
A–D	9	D–F	7
B–C	5	D–G	8
B–E	7	E–F	3
C–D	6	E–G	4
C–E	8	F–G	5

(a) Explain what is meant by a minimum spanning tree and why it is useful in network design. (4 Marks)

(b) Use either Prim's algorithm or Kruskal's algorithm to determine a minimum spanning tree for this network. Show the main steps of the algorithm. (10 Marks)

(c) State the selected links and the total installation cost of the spanning tree. (4 Marks)

(d) The firm now requires one additional backup link so that there is exactly one cycle in the final design. Identify the cheapest such addition and state the revised total cost. (4 Marks)

(e) Comment briefly on one limitation of using only an MST for a real logistics or telecoms network. (3 Marks)

Total 25 Marks

4.

Source: Practice Paper A, original Question 4.

Westmoor Cars is deciding how many display bays to provide for newly delivered vehicles. Each week, the number of cars delivered and the number of cars requested by customers are uncertain.

Cars Delivered	6	7	8	9	10
Probability	0.10	0.20	0.35	0.25	0.10

Weekly Demand	5	6	7	8	9
Probability	0.15	0.25	0.30	0.20	0.10

Each permanent display bay costs £18 per week. Any cars that cannot be placed in a display bay must be held in overflow storage at a cost of £30 per car per week.

For consistency, use the following random numbers for a 12-week simulation:

Week	1	2	3	4	5	6	7	8	9	10	11	12
Delivery RN	0.119	0.503	0.512	0.860	0.103	0.223	0.601	0.557	0.783	0.548	0.731	0.768
Demand RN	0.751	0.587	0.240	0.614	0.111	0.817	0.450	0.815	0.685	0.679	0.210	0.252

(a) Describe how you would model this problem as a simulation study. Your answer should identify the stochastic inputs, state variables and performance measures. (5 Marks)

(b) Simulate 12 weeks of operation for a capacity of 7 display bays using the random numbers above. You may use Excel or Python to carry out the calculations, but your logic must be clear. (8 Marks)

(c) Estimate the average weekly cost for a capacity of 7 bays. (3 Marks)

(d) Repeat the analysis for capacities of 6, 8 and 9 bays using the same random numbers, then recommend a capacity. Present your results in a clearly labelled table. (6 Marks)

(e) State two reasons why a much longer simulation or repeated runs would improve confidence in the recommendation. (3 Marks)

Total 25 Marks

5.

Source: Practice Paper A, original Question 5.

The demand analyst at Calder Kitchens has collected the following quarterly sales data for one of its product ranges.

Period	Year	Quarter	Demand
1	2020	Q1	510
2	2020	Q2	620
3	2020	Q3	570
4	2020	Q4	690
5	2021	Q1	548
6	2021	Q2	662
7	2021	Q3	608
8	2021	Q4	742
9	2022	Q1	590
10	2022	Q2	708
11	2022	Q3	646
12	2022	Q4	792
13	2023	Q1	636
14	2023	Q2	760
15	2023	Q3	692
16	2023	Q4	852

(a) Using periods 1–12 only, calculate suitable centred 4-quarter moving averages and estimate the trend where possible. Plot the original data together with the estimated trend. (8 Marks)

(b) Using periods 1–12 only, calculate seasonal indices using both additive and multiplicative decomposition. (7 Marks)

(c) Use periods 13–16 as a validation set. Produce forecasts for those periods using both models and compare their accuracy using one suitable error measure. (6 Marks)

(d) Using the better model, forecast the four quarters of 2024 and justify your model choice briefly. (4 Marks)

Total 25 Marks

6.

Source: Practice Paper A, original Question 6.

Northfield Relief ships medical supplies from three depots to four clinics through two regional hubs. Supplies may travel only from a depot to a hub and then from a hub to a clinic.

Depot	Supply	Cost to H_1	Cost to H_2
D_1	100	5	7
D_2	90	4	6
D_3	110	6	3

Hub	Cost to C_1	Cost to C_2	Cost to C_3	Cost to C_4	Total Demand
H_1	4	6	5	7	
H_2	6	4	3	5	
Demand	60	70	80	75	285

(a) Formulate a transhipment model for this problem. Clearly define the decision variables, objective function, supply constraints, demand constraints and hub flow-balance constraints. (8 Marks)

(b) Solve the model and determine the optimal shipment plan, the total transportation cost and any unused supply. Report only the non-zero shipments. (7 Marks)

(c) A disruption means that hub H_1 can process at most 130 units, and the route from H_2 to clinic C_3 is unavailable. Show how the model changes and determine the revised total cost. State only the revised non-zero shipments or the flows that change. (6 Marks)

(d) In the disrupted scenario, up to 20 extra units of processing capacity can be bought for hub H_1 at a one-off cost of £50. Determine whether this is worthwhile and justify your answer. (4 Marks)

Total 25 Marks

7.

Source: Practice Paper B, original Question 1.

Silverforge Engineering is studying two scheduling problems.

Part A: Flow Shop Scheduling

All five jobs must be processed first on Machine 1 and then on Machine 2.

Job	Machine 1 Time	Machine 2 Time
J_1	7	5
J_2	3	8
J_3	6	4
J_4	2	9
J_5	5	6

Part B: Job Shop Scheduling

Each job has its own machine order and must follow that order exactly.

Job	Sequence on Machines	Processing Times
A	M_1, M_3, M_2	$4, 5, 3$
B	M_2, M_1, M_3	$3, 4, 6$
C	M_3, M_2, M_1	$5, 2, 4$
D	M_1, M_2, M_3	$2, 6, 3$

(a) Use Johnson's Rule to determine an optimal production sequence for the flow shop problem in Part A. Show the main steps. (6 Marks)

(b) Draw a Gantt chart for your sequence and calculate the makespan. (5 Marks)

(c) Compare your Johnson schedule with the original order J_1, J_2, J_3, J_4, J_5 and comment on the improvement. (4 Marks)

(d) Solve the job shop scheduling problem in Part B and state an optimal schedule and the corresponding makespan. You may present the result as a table of operation start and finish times or as a Gantt chart. (6 Marks)

(e) Suppose job C must finish by time 15. Explain how the model changes and discuss the possible effect on feasibility and makespan. (4 Marks)

Total 25 Marks

8.

Source: Practice Paper B, original Question 2.

County Response wants to choose a limited number of patrol bases. The aim is not to cover every district, but to maximise the total population covered while opening at most three bases.

Base	Cost	Population Covered	Districts Covered
A	14	18,000	$\{1, 2, 4\}$
B	11	15,000	$\{1, 3\}$
C	13	21,000	$\{2, 3, 5\}$
D	10	12,000	$\{4, 5\}$
E	12	16,000	$\{2, 5, 6\}$
F	9	14,000	$\{3, 4, 6\}$

The population living in each district is given below.

District	1	2	3	4	5	6
Population	9,000	12,000	14,000	10,000	11,000	8,000

The budget is £31 and at most 3 bases may be opened.

(a) Formulate a binary integer programming model to maximise covered population subject to the budget and base limit. Clearly define any coverage variables you use. (9 Marks)

(b) Solve the model and identify which bases should be opened and the resulting total covered population. (6 Marks)

(c) Explain whether some districts may remain uncovered in an optimal solution and why this is acceptable in this model but not in a set covering model. (4 Marks)

(d) Modify the model so that district 6 must be covered and bases B and F cannot both be chosen. State the revised model changes clearly. (6 Marks)

Total 25 Marks

9.

Source: Practice Paper B, original Question 3.

A university library help desk is modelled as a queueing system. Students arrive randomly and service times are assumed to follow an exponential distribution. You should adapt the SimPy structures used in class, but tailor the model to this setting.

Use the following assumptions:

Quantity	Value
Simulation length	360 minutes
Average service time	7 minutes
Baseline arrival rate	10 students per hour
Peak arrival rate	15 students per hour
Service target	97% wait less than 8 minutes
Priority users	10% of arrivals
Suggested random seed	24

(a) Describe the main elements of a discrete-event simulation model for this system: entities, resources, events, stochastic inputs and performance measures. (5 Marks)

(b) Adapt a SimPy model to represent the baseline system with 2 advisers and an arrival rate of 10 students per hour. Report the total served, average wait, maximum wait and percentage waiting less than 8 minutes. (6 Marks)

(c) Investigate the peak-period scenario with 2, 3 and 4 advisers. Present your results in a clearly labelled table and determine the minimum number of advisers needed to meet the service target. (4 Marks)

(d) Extend the model so that 10% of users are priority users who should be served ahead of standard users. Explain how the model should be changed and discuss the likely effect on standard-user waits. (6 Marks)

(e) Explain why a single run is not enough to support a staffing decision and describe how you would use replications and/or a warm-up period to improve the analysis. (4 Marks)

Total 25 Marks

10.

Source: Practice Paper B, original Question 4.

The operations analyst at Broadleaf Homeware has collected the following quarterly demand data.

Period	Year	Quarter	Demand
1	2019	Q1	420
2	2019	Q2	500
3	2019	Q3	470
4	2019	Q4	560
5	2020	Q1	450
6	2020	Q2	536
7	2020	Q3	500
8	2020	Q4	600
9	2021	Q1	485
10	2021	Q2	575
11	2021	Q3	535
12	2021	Q4	645
13	2022	Q1	522
14	2022	Q2	618
15	2022	Q3	572
16	2022	Q4	694

(a) Using periods 1–12 only, calculate suitable centred 4-quarter moving averages and estimate the trend values where possible. Plot the original data together with the estimated trend. (8 Marks)

(b) Using periods 1–12 only, calculate seasonal indices using both additive and multiplicative decomposition models. (7 Marks)

(c) Use the 2022 data (periods 13–16) as a validation set. Generate forecasts for these four periods using both models and compare their accuracy using one suitable measure such as MAD, MSE or MAPE. (6 Marks)

(d) Using the better model, produce forecasts for the four quarters of 2023 and justify your final model choice. (4 Marks)

Total 25 Marks

11.

Source: Practice Paper B, original Question 5.

Harbour Plastics makes two products each week: casings and inserts.

Each casing requires 4 moulding hours and 1 assembly hour. Each insert requires 2 moulding hours and 3 assembly hours. At most 100 moulding hours and 90 assembly hours are available. No more than 24 inserts can be sold each week.

The weekly contribution is £40 per casing and £34 per insert.

(a) Formulate a linear programming model for this problem. (6 Marks)

(b) Solve the model to determine the production plan that maximises weekly contribution. (6 Marks)

(c) State the binding constraints and the unused capacity in each department. (4 Marks)

(d) A distributor now requires at least 25% of total output to be inserts. Reformulate the model and determine whether the original solution remains feasible. (5 Marks)

(e) If up to 12 extra assembly hours can be purchased at £3 per hour for one week only, determine whether they should be bought. (4 Marks)

Total 25 Marks

12.

Source: Practice Paper B, original Question 6.

Northline Distribution must ship goods from three warehouses to four stores at minimum cost.

Warehouse	S_1	S_2	S_3	S_4
W_1	6	8	10	7
W_2	9	5	7	6
W_3	4	7	6	8

Warehouse	Supply
W_1	80
W_2	100
W_3	90

Store	Demand
S_1	70
S_2	60
S_3	80
S_4	60

(a) Formulate a transportation model for this problem, clearly defining the decision variables, objective function and constraints. (7 Marks)

(b) Solve the model and determine the optimal shipping plan and total transportation cost. Report only the non-zero shipments. (7 Marks)

(c) Resolve the model if warehouse W_2 cannot supply store S_2, and at least 20 units must be shipped from W_1 to S_4. Comment on the impact on total cost. (6 Marks)

(d) Explain one realistic extension that would make this transportation model more useful in practice. (5 Marks)

Total 25 Marks